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Foot and ankle compensation for anterior cruciate ligament deficiency during gait in children

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ABSTRACT

Background: Anterior cruciate ligament (ACL) injuries are common in adults and cause knee instability, pain, and an increased risk of osteoarthritis. Previous studies demonstrated changed gait patterns in adult patients with ACL deficiency. In paediatric patients, ACL injuries were once thought to be rare but are being increasingly diagnosed due to greater involvement of children in contact sports and to the introduction of more effective diagnostic tools such as magnetic resonance imaging (MRI). However, little is known about gait adaptation in children with ACL deficiency. The objective of this study was to look for compensatory foot and ankle behaviours during gait in paediatric patients with symptomatic ACL deficiency.
Hypothesis: Compensation for ACL deficiency during gait occurs at the foot and ankle in children, because compensation at the hip and pelvis would require greater energy expenditure.
Material and methods: We included 47 patients, 33 males and 14 females, ranging in age from 9 to 17 years (mean, 14.1 years). The patients had a history of unilateral ACL injury documented by MRI and initially treated by immobilisation and physical therapy. They were allowed to walk with full weight-bearing on the affected limb and were not taking medications at the time of the study. All patients had pain, knee instability, or functional limitation. The physical examination showed joint laxity indicating surgical ACL reconstruction. None had neurological conditions, congenital musculoskeletal abnormalities, or a history of knee surgery. Gait analysis (GA) was performed using a Vicon 460 system. Kinematic data for the ankle and foot were compared to those in a control group of 37 healthy children. Ankle angular positions were calculated for each group at the following stance time points: initial contact (0% of gait cycle [GC]), mid-stance (25% GC), terminal stance (60% GC), and swing (83% GC). Foot progression data were recorded at mid-stance (25% GC) and swing (70% GC). Student's *t* test was applied to compare the results to reference values obtained at our laboratory and to data from the control group.
Results: Compared to the reference values, the ankle was in plantar flexion at initial contact in 41 patients, and ankle dorsiflexion during the stance phase was diminished in 39 patients. The external foot progression angle was increased in 23 patients during the stance phase and 38 patients during the swing phase. Compared to the control group (mean age, 9.1 years), the patients had plantar flexion of the ankle at initial contact ($3.43° \pm 3.5°$ vs. $0.74° \pm 3.6°$, $p < 0.05$) and decreased dorsiflexion during the stance phase ($3.43° \pm 3.5°$ vs. $0.74° \pm 3.6°$, $p < 0.05$). No significant differences were found for any of the other parameters.
Discussion: Children with ACL deficiency developed compensatory foot and ankle behaviours during gait that improved knee stability. Understanding these compensations may guide treatment optimisation.
Level of evidence: III, retrospective comparative study.

1. Introduction

Anterior cruciate ligament (ACL) tears are common injuries that result in knee instability and increase the risk of early osteoarthritis [1,2]. Several studies in adults with a history of ACL injury

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documented changes in gait involving walking speed and spatiotemporal parameters, with decreased knee motion ranges that persisted even after ligament reconstruction [1,3]. These changes reflect compensatory muscular behaviours that limit anterior tibial translation and improve knee stability. Bulgheroni et al. reported that gait parameters improved after ACL reconstruction [4].

ACL injuries have a rising incidence in paediatric patients [5] due to the growing popularity among children of contact sports associated with severe knee injuries. In addition, advances in imaging techniques, notably magnetic resonance imaging (MRI), have improved the diagnosis of injuries to the central knee structures. Parkkari et al. [6] reported 265 cruciate ligament injuries (with 92% of ACL tears) in a Finnish population-based cohort study of 46 472 adolescents, yielding an incidence of 60.9 per 100 000 person-years. Concomitant injuries to the menisci and collateral ligaments are common and further increase the risk of early-onset osteoarthritis [7,8]. Werner et al. [9] found a significant increase in diagnosis of ACL injury and ACL reconstruction in paediatric and adolescent patients when compared to adults between 2007 and 2011. The results of several studies support early ACL reconstruction in children to protect the menisci, restore normal knee motion ranges, and allow the return to sports [7–14].

Although they have increased capacity of muscle adaptation and gait compensation, children with ACL injuries complain of unstable knee and difficulties during walking. Most studies rely on radiological data to assess outcomes, and little information is available on compensatory behaviours during gait.

The objective of this study was to look for foot and ankle adaptations during gait in paediatric patients with symptomatic ACL deficiency. The working hypothesis was that compensation for ACL deficiency during gait occurs at the foot and ankle in children, because compensation at the hip and pelvis would require greater energy expenditure.

2. Material and method

A retrospective study was conducted in 47 patients, 33 males and 14 females, ranging in age from 9 to 17 years (mean, 14.0 ± 1.9 years). All patients had a history of unilateral ACL injury documented by MRI. The injury occurred during sports in 37 (78.7%) patients and a traffic accident in 2 (4.2%) patients; no data on the circumstances of the injury were available for the remaining 8 patients. The left knee was affected in 28 patients and the right knee in 19 patients. The mean age at injury was 12.0 ± 2.9 years. All patients initially received non-operative treatment consisting of immobilisation followed by specialised rehabilitation exercises including quadriceps strengthening and proprioception training. Although the rehabilitation programme was optimally adapted to the characteristics of the injury, all patients remained symptomatic and had persistent knee instability that precluded their return to sports.

At the time of the study, all patients walked with full weight-bearing on the affected side, and none required analgesics. The physical examination and knee laxity measurements confirmed the instability. The paediatric orthopaedic surgeons managing the patients deemed that surgery was necessary based on clinical criteria (knee instability, pain, and functional impairments). Gait analysis (GA) was performed before surgery. The time from injury to GA ranged from 2 to 48 months (mean, 13 months).

Gait analysis was performed using the Vicon 360 motion capture system (Vicon Motion Systems, Oxford, UK). The kinematic data were recorded by five 60-Hz infrared cameras from 15 reflective markers positioned according to the Helen Hayes model. The patients walked at their usual speed along a 10-metre distance, and 16 gait cycles (GCs) were recorded. Data analysis was with Vicon Polygon software. The following spatiotemporal parameters were recorded: speed, cadence, stride and step length, and percentages of time spent in single-leg and double-leg stance. Kinematic data on the ankle and foot in the sagittal plane were compared to those obtained in a control group of 37 children free of orthopaedic abnormalities, 21 females and 6 males, ranging in age from 3 to 15 years (mean, 9 ± 3 years). Ankle flexion and extension were computed at the following points of the GC: initial contact (0%), mid-stance (25%), terminal stance (60%), and swing phase (83%). The foot progression angle was determined at mid-stance (25%) and during the swing phase (70%).

Statistical comparisons were with Student's *t* test. Values of p below 0.05 were considered significant.

3. Results

None of the spatiotemporal parameters differed significantly between the patients and the controls. In contrast, the group of patients exhibited modifications in ankle kinematics that predominated at the early stance phase (Table 1). At initial contact (0% of the GC), the ankle was in plantar flexion (0.2° to 15.1°) in 41 patients and in dorsal flexion (0.1° to 2.6°) in 6 patients. Mean ankle position at initial contact in the ACL group was $3.4^\circ \pm 3.5^\circ$ of plantar flexion. In the control group, mean ankle position at initial contact was $0.74^\circ \pm 3.6^\circ$ of dorsiflexion ($p = 2.5510^{-9}$). At mid-stance (25% of the GC), compared to the controls, ankle dorsiflexion was diminished in 41 patients and increased in 6 patients. Mean dorsiflexion was $5.9^\circ \pm 2.1^\circ$ in the ACL group and $7.9^\circ \pm 2.5^\circ$ in the controls ($p = 1.28\ 10^{-11}$) (Fig. 1). At the terminal stance phase (60% of the GC), 26 patients had increased plantar flexion compared to the controls (mean, $14.1^\circ \pm 6.3^\circ$ vs. $12.9^\circ \pm 6.6^\circ$, respectively), but the difference was not statistically significant ($p > 0.05$). Finally, during the swing phase 22 ACL patients had less ankle dorsiflexion compared to the controls (mean, $0.7^\circ \pm 2.7^\circ$ vs. $1.0^\circ \pm 3.4^\circ$, respectively), but again the difference was not statistically significant.

The comparison of the foot progression angle showed greater external rotation in the patient group than in the control group. However, the difference was not statistically significant ($p > 0.05$) (Table 2).

4. Discussion

This study demonstrated that ACL deficiency was associated with abnormal ankle kinematics. Ankle dorsiflexion was decreased at the initial contact and early stance phases, during which the differences with the control group were statistically significant. These findings confirm our working hypothesis. The alterations in ankle position in our patients suggest that, when the ACL is deficient, plantar flexion of the ankle may facilitate knee extension. For biomechanical reasons, plantar flexion of the ankle at initial contact results in passive knee extension because the ground reaction force vector is anterior to the knee (with an external knee extension moment). This extension of the knee does not require contraction of the quadriceps, which might translate the tibia anteriorly. Hicks et al. [15] investigated joint stabilisation by the muscles that can modify the position of the joints they do not cross and showed, in particular, that the soleus muscle is involved in coupling plantar flexion of the ankle and extension of the knee.

Our assessment of the foot progression angle showed increased external rotation, which was most marked during the swing phase, although the difference was not statistically significant (Fig. 2). Similarly, Zhang et al. [16] reported increased external rotation of the tibia during gait in patients with anterior knee instability. During the stance phase, external rotation of the foot results in lateral and posterior displacement of the ground reaction force compared to

Table 1
Ankle dorsiflexion during the stance and swing phases of the gait cycle in the patients with ACL deficiency and the healthy controls.

Ankle dorsiflexion,°, mean ± SD	Patients (n = 47)	Controls (n = 37)	p value
0% of gait cycle	−3.44 ± 3.54	0.74 ± 3.67	$2.55 \cdot 10^{-9}$
25% of gait cycle	5.91 ± 2.15	7.96 ± 2.52	$1.28 \cdot 10^{-11}$
60% of gait cycle	−14.18 ± 6.40	−12.99 ± 6.60	> 0.05
83% of gait cycle	0.78 ± 2.72	1.09 ± 3.43	> 0.05

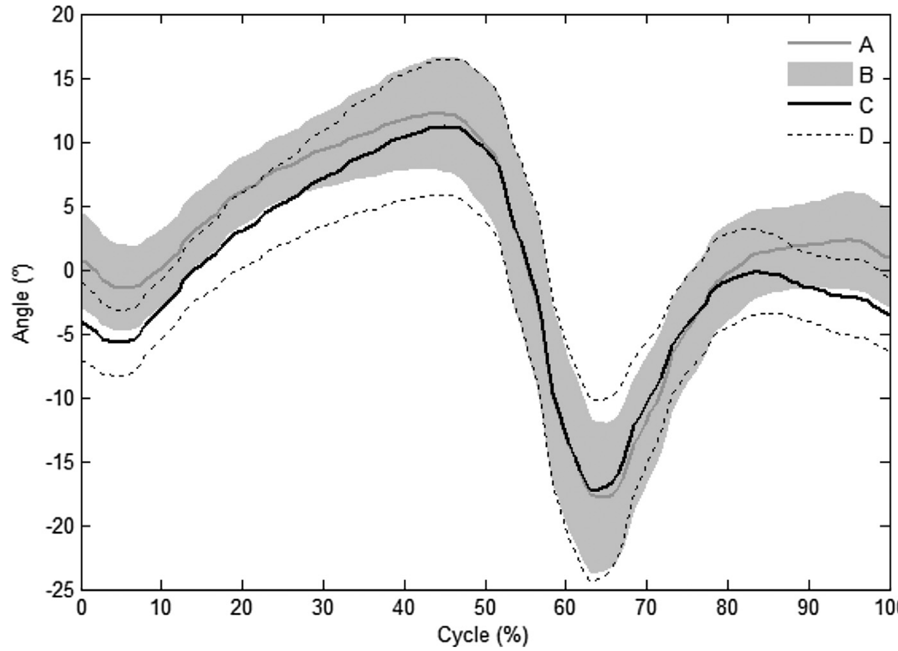

Fig. 1. Ankle kinematics. A. mean value in the control group; B. individual values in the control group; C. mean value in the group of patients with ACL deficiency; D. individual values in the group of patients with ACL deficiency.

Table 2
External foot progression angle during the stance and swing phases of the gait cycle in the patients with ACL deficiency and the healthy controls.

Plantar progression,°, mean ± SD	Patients (n = 47)	Controls (n = 37)	p value
25% of the gait cycle	−5.2 ± 4.81	−6.32 ± 5.5	> 0.05
70% of the gait cycle	−12.9 ± 7.1	−10.36 ± 6.59	> 0.05

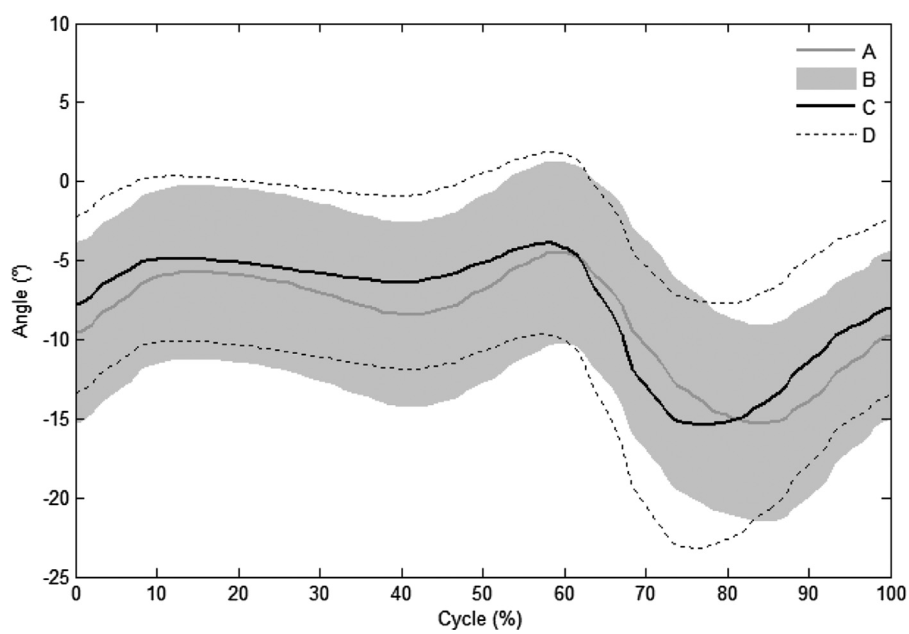

Fig. 2. Foot progression angle. A. mean value in the control group; B. individual values in the control group; C. mean value in the group of patients with ACL deficiency; D. individual values in the group of patients with ACL deficiency.

its normal position. The effect is shortening of the lever arm with a decrease in the extension moment of the knee.

Our institution is in an area where contact sports, notably rugby, are popular among the paediatric population, notably boys. This fact may explain the 3:1 male-to-female ratio in our study, which contrasts with several reports that females are at greater risk for ACL injury [14,17,18]. Furthermore, in an MRI study by Prince et al. [19], ACL tears were common before skeletal maturity in boys and after growth completion in girls.

Several studies have documented gait alterations in patients with ACL deficiency [4,5,18]. Berchuck et al. [20] analysed the abnormalities found during walking, jogging, and ascending and descending stairs in 16 patients with unilateral ACL injuries and in 10 healthy controls. In the patients, even mild physical activity such as walking was associated with a decrease in the knee flexion moment, which avoided contraction of the quadriceps. Contraction of the quadriceps with the knee between 0° and 40° of flexion can cause anterior translation of the tibia in the ACL-deficient knee. The knee extension moment at initial contact was significantly increased in the patients compared to the controls. Ankle kinematics were not significantly different between the two groups.

In 18 patients with anterior knee instability, Roberts et al. [21] found no reduction in the internal knee extension moment. They concluded from their findings that the quadriceps avoidance pattern as a gait adaptation to ACL deficiency may be less common than previously reported.

The ACL is among the knee stabilisers, as it prevents anterior translation of the tibia. In addition, during gait, the muscles provide dynamic stabilisation of the knee. Coordinated activity of the quadriceps and hamstrings is crucial to good knee mobility. For biomechanical reasons, contraction of the quadriceps promotes anterior translation of the proximal tibia, whereas the ACL and contraction of the hamstrings have the opposite effect [22–25]. Several studies of the part played by dynamic muscle control identified the gastrocnemius muscles as key knee stabilisers. Using electromyography, changes in the intensity and timing of gastrocnemius activity have been documented in adults with ACL injuries [25–27].

Our study focused on ankle and foot kinematics. The Gait analysis results at the hip and knee were not significantly different between the patients and controls. We are currently performing a larger study to investigate the relations linking contraction of the hamstrings and quadriceps and also of the hamstrings and triceps surae. with narrower age groups to allow for the identification of age-specific morphological and functional characteristics.

To our knowledge, this is the first study of gait alterations designed to compensate for ACL deficiency in children. Consequently, there are no data to which our findings can be compared. We confined our study to symptomatic patients who had functional impairments and were scheduled for surgical stabilisation. In the future, studies of patients who have mild symptoms or normal function should be performed.

5. Conclusion

Our findings confirm the hypothesis that children alter their gait to compensate for ACL deficiency. The alterations differed from those reported in adults. In particular, plantar flexion of the ankle at initial contact was noted. Compensatory behaviours during gait may be difficult to detect during the physical examination. Gait analysis is a reliable tool for elucidating compensatory mechanisms and optimising rehabilitation therapy.

Disclosure of interest

The authors declare that they have no competing interest.

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Contributions of each author

All authors contributed equally to this work.

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